



## **Victorian Certificate of Education 2013**

# **PHYSICS**

## **Written examination**

**Wednesday 13 November 2013**

**Reading time: 9.00 am to 9.15 am (15 minutes)**

**Writing time: 9.15 am to 11.45 am (2 hours 30 minutes)**

## **FORMULA SHEET**

### **Directions to students**

- A question and answer book is provided with this formula sheet.

**Students are NOT permitted to bring mobile phones and/or any other unauthorised electronic devices into the examination room.**

|    |                                                          |                                                                                                                 |
|----|----------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------|
| 1  | velocity; acceleration                                   | $v = \frac{\Delta x}{\Delta t}; \quad a = \frac{\Delta v}{\Delta t}$                                            |
| 2  | equations for constant acceleration                      | $v = u + at$ $x = ut + \frac{1}{2}at^2$ $v^2 = u^2 + 2ax$ $x = \frac{1}{2}(v + u)t$                             |
| 3  | Newton's second law                                      | $\Sigma F = ma$                                                                                                 |
| 4  | circular motion                                          | $a = \frac{v^2}{r} = \frac{4\pi^2 r}{T^2}$                                                                      |
| 5  | Hooke's law                                              | $F = -kx$                                                                                                       |
| 6  | elastic potential energy                                 | $\frac{1}{2}kx^2$                                                                                               |
| 7  | gravitational potential energy near the surface of Earth | $mgh$                                                                                                           |
| 8  | kinetic energy                                           | $\frac{1}{2}mv^2$                                                                                               |
| 9  | Newton's law of universal gravitation                    | $F = G \frac{M_1 M_2}{r^2}$                                                                                     |
| 10 | gravitational field                                      | $g = G \frac{M}{r^2}$                                                                                           |
| 11 | acceleration due to gravity at Earth's surface           | $g = 10 \text{ m s}^{-2}$                                                                                       |
| 12 | voltage; power                                           | $V = RI \quad P = VI = I^2 R$                                                                                   |
| 13 | resistors in series                                      | $R_T = R_1 + R_2$                                                                                               |
| 14 | resistors in parallel                                    | $\frac{1}{R_T} = \frac{1}{R_1} + \frac{1}{R_2}$                                                                 |
| 15 | transformer action                                       | $\frac{V_1}{V_2} = \frac{N_1}{N_2}$                                                                             |
| 16 | AC voltage and current                                   | $V_{\text{RMS}} = \frac{1}{\sqrt{2}} V_{\text{peak}} \quad I_{\text{RMS}} = \frac{1}{\sqrt{2}} I_{\text{peak}}$ |
| 17 | magnetic force                                           | $F = I l B$                                                                                                     |

|    |                                                                    |                                                                                                                                                   |
|----|--------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------|
| 18 | electromagnetic induction                                          | emf : $\varepsilon = -N \frac{\Delta \Phi}{\Delta t}$ flux: $\Phi = BA$                                                                           |
| 19 | transmission losses                                                | $V_{\text{drop}} = I_{\text{line}} R_{\text{line}}$ $P_{\text{loss}} = I_{\text{line}}^2 R_{\text{line}}$                                         |
| 20 | mass of the electron                                               | $m_e = 9.1 \times 10^{-31} \text{ kg}$                                                                                                            |
| 21 | charge on the electron                                             | $e = -1.6 \times 10^{-19} \text{ C}$                                                                                                              |
| 22 | Planck's constant                                                  | $h = 6.63 \times 10^{-34} \text{ J s}$<br>$h = 4.14 \times 10^{-15} \text{ eV s}$                                                                 |
| 23 | speed of light                                                     | $c = 3.0 \times 10^8 \text{ m s}^{-1}$                                                                                                            |
| 24 | photoelectric effect                                               | $E_{K\text{max}} = hf - W$                                                                                                                        |
| 25 | photon energy                                                      | $E = hf$                                                                                                                                          |
| 26 | photon momentum                                                    | $p = \frac{h}{\lambda}$                                                                                                                           |
| 27 | de Broglie wavelength                                              | $\lambda = \frac{h}{p}$                                                                                                                           |
| 28 | speed, frequency and wavelength                                    | $v = f\lambda$                                                                                                                                    |
| 29 | energy transformations for electrons in an electron gun (<100 keV) | $\frac{1}{2}mv^2 = eV$                                                                                                                            |
| 30 | radius of electron path                                            | $r = \frac{mV}{eB}$                                                                                                                               |
| 31 | magnetic force on a moving electron                                | $F = evB$                                                                                                                                         |
| 32 | Bragg's law                                                        | $n\lambda = 2d\sin \theta$                                                                                                                        |
| 33 | electric field between charged plates                              | $E = \frac{V}{d}$                                                                                                                                 |
| 34 | band gap energy                                                    | $E = \frac{hc}{\lambda}$                                                                                                                          |
| 35 | Snell's law                                                        | $n_1 \sin \theta_1 = n_2 \sin \theta_2$                                                                                                           |
| 36 | intensity and level                                                | sound intensity level (in dB)<br>$L(\text{dB}) = 10 \log_{10} \left( \frac{I}{I_0} \right)$<br>where $I_0 = 1.0 \times 10^{-12} \text{ W m}^{-2}$ |

|    |                                  |                                                          |
|----|----------------------------------|----------------------------------------------------------|
| 37 | Lorentz factor                   | $\gamma = \frac{1}{\sqrt{1 - \frac{v^2}{c^2}}}$          |
| 38 | time dilation                    | $t = t_0 \gamma$                                         |
| 39 | length contraction               | $L = \frac{L_0}{\gamma}$                                 |
| 40 | relativistic mass                | $m = m_0 \gamma$                                         |
| 41 | total energy                     | $E_{\text{total}} = E_k + E_{\text{rest}} = mc^2$        |
| 42 | stress                           | $\sigma = \frac{F}{A}$                                   |
| 43 | strain                           | $\varepsilon = \frac{\Delta L}{L}$                       |
| 44 | Young's modulus                  | $E = \frac{\text{stress}}{\text{strain}}$                |
| 45 | capacitors                       | time constant : $\tau = RC$                              |
| 46 | universal gravitational constant | $G = 6.67 \times 10^{-11} \text{ N m}^2 \text{ kg}^{-2}$ |
| 47 | mass of Earth                    | $M_E = 5.98 \times 10^{24} \text{ kg}$                   |
| 48 | radius of Earth                  | $R_E = 6.37 \times 10^6 \text{ m}$                       |
| 49 | mass of the electron             | $m_e = 9.1 \times 10^{-31} \text{ kg}$                   |
| 50 | charge on the electron           | $e = -1.6 \times 10^{-19} \text{ C}$                     |
| 51 | speed of light                   | $c = 3.0 \times 10^8 \text{ m s}^{-1}$                   |

**Prefixes/Units**p = pico =  $10^{-12}$ n = nano =  $10^{-9}$  $\mu$  = micro =  $10^{-6}$ m = milli =  $10^{-3}$ k = kilo =  $10^3$ M = mega =  $10^6$ G = giga =  $10^9$ t = tonne =  $10^3 \text{ kg}$ **END OF FORMULA SHEET**